

- By Defn. of OR, $x \in A \vee x \in \bar{A}$ is True.
- By Defn. of union, $x \in A \cup \bar{A}$.

g) Prove the complement laws in Table 1 by showing that

a) $A \cup \bar{A} = U$ b) $A \cap \bar{A} = \emptyset$

Solution: a) Let, $x \in A \cup \bar{A}$. x is an arbitrary element.
By Defn. 1, $x \in A \vee x \in \bar{A}$. By Defn. of OR, at least one of $x \in A$ or $x \in \bar{A}$ is True.

Let, $x \in A$. By Defn. of negation, $\neg(x \in A)$ is False. By Defn. of not belongs to, $x \notin A$ is False. By Defn. 5, $x \in \bar{A}$ is False.
We know that $A \subseteq U$. By Defn. 3 of Sec-2.1, $x \in U$. By Defn. 3 of Section 2.1, $A \cup \bar{A} \subseteq U$.

Let $x \in \bar{A}$. By Defn. 5, $x \notin A$. By Defn. of not belongs to, $x \in U - A$. By Defn. 4, $x \in U$ but $x \notin A$. By Defn. of not belongs to, $\neg(x \in A)$ is True. By Defn. of negation, $\neg(\neg(x \in A))$ is False. By Double-negation Law of propositions, $x \in A$ is False. By Defn. 3 of Section 2.1, $A \cup \bar{A} \subseteq U$.

Let $x \in U$. ~~Let~~ Let, $x \in A$. x is an arbitrary element of U .
By Defn. of negation, $\neg(x \in A)$ is False. By Defn. of not belongs to, $x \notin A$ is False.
By Defn. 5, $x \in \bar{A}$ is False. By Defn. of OR, $x \in A \vee x \in \bar{A}$ is True. By Defn. 1, $x \in A \cup \bar{A}$.
Let $x \in \bar{A}$. By Defn. 5, $x \notin A$. By Defn. of not belongs to, $\neg(x \in A)$ is True.
By Defn. of negation, $\neg(\neg(x \in A))$ is False. By Double-negation Law of propositions, $x \in A$ is False. By Defn. of OR, $x \in A \vee x \in \bar{A}$ is True. By Defn. 1, $x \in A \cup \bar{A}$.
By Defn. 3 of Section 2.1, $U \subseteq A \cup \bar{A}$.

$\therefore A \cup \bar{A} = U$ $\square$

b) Let, x be an arbitrary element of $A \cap \bar{A}$, i.e. $x \in A \cap \bar{A}$.
By Defn. 2, $x \in A \wedge x \in \bar{A}$. By Defn. of AND, both $x \in A$ and $x \in \bar{A}$.
By Defn. 5, $x \notin A$. This is a contradiction, i.e. $x \in \emptyset$. By Defn. 3 of Section 2.1, $A \cap \bar{A} \subseteq \emptyset$.